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Module 7 Introduction

Lesson 1: Foundations of Bayesian Regression Analysis

Lesson 2: Advanced Techniques in Hierarchical Linear Models

Lesson 3: Exploring Generalized Linear Models in Bayesian Context

Reading: Exploring Generalized Linear Models in Bayesian Context Readings 1h

Video: Generalized Linear Models Pt. 1 1 min

Video: Generalized Linear Models Pt. 2 2 min

Video: Generalized Linear Models Pt. 3 1 min

Video: Generalized Linear Models Pt. 4 3 min

Video: Generalized Linear Models Pt. 5 3 min

Video: Generalized Linear Models Pt. 6 6 min

Video: Generalized Linear Models Pt. 7 8 min

Practice Assignment: Exploring Generalized Linear Models in Bayesian Context Quiz Submitted

Lab: Module 7 - Lesson 3 - RStudio Lab 1h

Module 7 Summative Assessment

Module 7 Summary

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass, you need at least 80%. We keep your highest score.

Next item →

Exploring Generalized Linear Models in Bayesian Context Quiz

Instructions

Review Learning Objectives

1. What is a generalized linear model used for? 1 / 1 point

- ☐ To model linear relationships only

☐ To model relationships where the normal distribution is always appropriate

☐ To simplify linear regression

☒ To unify approaches for analyzing data with non-normal and non-linear relationships

Assignment details

Submitted

Attempts

Unlimited

Retry

Correct

Correct! Generalized linear models unify approaches for analyzing data where linear or normal assumptions are not appropriate. To pass, you need at least 80%. We keep your highest score.

View submission

See feedback

100%

2. What is the role of the link function in a generalized linear model? 1 / 1 point

- ☐ It relates the explanatory variables directly to the outcome variable

☐ It transforms the explanatory variables

☒ It relates the linear predictor to the mean of the outcome variable

☐ It determines the distribution of the residuals

Correct

Correct! The link function $g(\cdot)$ relates the linear predictor $\eta = X\beta$ to the mean μ of the outcome variable.

3. What is a common link function used in Poisson regression? 1 / 1 point

- ☐ Identity link

☐ Probit link

☐ Logit link

☒ Log link

Correct

Correct! The log link function is commonly used in Poisson regression.

